FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Main)-2020

TEST DATE: 29-12-2019

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A man standing on an inclined plane of inclination 'θ' with the horizontal, fires one bullet up the plane and another bullet down the plane. Both the bullets get stuck to the plane after hitting the inclined plane. The value of maximum possible distance between the two bullets after they hit the inclined plane is:

(A) $\frac{2u^2}{g}$

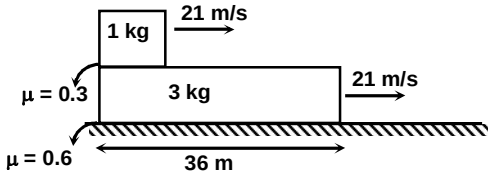
(B) $\frac{2u^2}{g\cos^2\theta}$

(C) $\frac{u^2}{g}$

(D) $\frac{u^2}{g\cos^2\theta}$

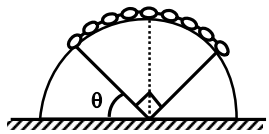
2. In Young's double slit experiment, if we increase the separation between the plane of slits and the screen by keeping everything constant, then
(A) Intensity of central maxima will increase but fringe width will decrease
(B) Intensity of central maxima will decrease but fringe width will increase.
(C) Both intensity of central maxima and fringe width will increase
(D) Both intensity of central maxima and fringe width will decrease.

3. On a plank of length 36 m and mass 3 kg, a small block of mass 1 kg is kept as shown. Friction coefficient between the block and the plank is 0.3 where as that between the plank and the ground is 0.6. At some instant an impulse is given to both of them such that they acquire a common initial velocity of 21 m/s as shown. After what time will the block get off the plank?



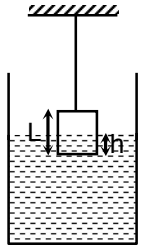
(A) $3\sqrt{2}$ sec
(B) 4 sec
(C) 5 sec
(D) $4\sqrt{2}$ sec

4. On a fixed hemispherical wedge a uniform chain is in equilibrium. Length of the chain is $\frac{\pi R}{2}$, where 'R' is the radius of the fixed wedge. Find the minimum value of 'θ' as shown in the figure. (take $\mu = 1/3$)



(A) $\tan^{-1}(1)$
(B) $\tan^{-1}(1/2)$
(C) $\tan^{-1}(1/3)$
(D) $\tan^{-1}(1/4)$

5. A cube (edge length L) of specific gravity 'n' is suspended from a thin metal wire. The fundamental frequency for transverse standing waves in the wire is f_0. When the same cube is partially immersed in a liquid of specific gravity 2 upto a depth of 'h', the new fundamental frequency will be



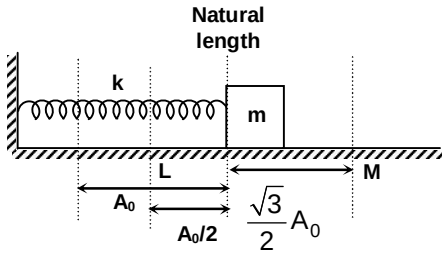
(A) $\left(\frac{nL-2h}{L}\right)^{1/2} f_0$

(B) $\left(\frac{nL-2h}{nL}\right)^{1/2} f_0$

(C) $\left(\frac{L}{nL-2h}\right)^{1/2} f_0$

(D) $\left(\frac{nL}{nL-2h}\right)^{1/2} f_0$

6. From a spring of spring constant 'k', a small block of mass 'm' is attached and the system is placed on a smooth horizontal surface as shown in the figure. The spring is compressed by distance A_0 from its natural length and then released. Find the time taken by the block to move from point 'L' to point 'M'.



(A) $\frac{\pi}{4}\sqrt{\frac{m}{k}}$

(B) $\frac{\pi}{3}\sqrt{\frac{m}{k}}$

(C) $\frac{\pi}{2}\sqrt{\frac{m}{k}}$

(D) $\frac{\pi}{6}\sqrt{\frac{m}{k}}$

7. Two thin rods of length L and 2L at certain temperature have coefficient of linear expansion 2α and α respectively. Both are joined end to end. Then, the effective coefficient of linear expansion of the composite rod is

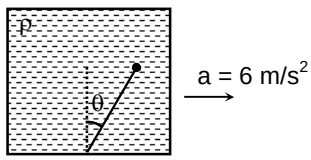
(A) 3α

(B) 4α

(C) $\frac{2}{3}\alpha$

(D) $\frac{4}{3}\alpha$

8. In an ideal liquid of density 'ρ', a small spherical ball of density $\frac{\rho}{5}$ is tied to a string which is connected to the bottom of a container as shown. If the container is having a horizontal acceleration of 6 m/s^2, then the angle made by the string with the vertical is: (Take g = 10 m/s^2)



(A) $\tan^{-1}\left(\frac{3}{5}\right)$

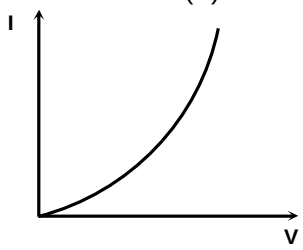
(B) $\tan^{-1}\left(\frac{4}{5}\right)$

(C) $\frac{\pi}{4}$

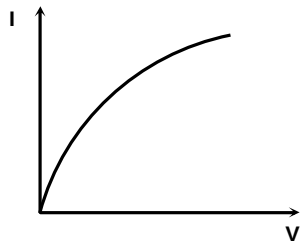
(D) $\tan^{-1}(5)$

9. Drift velocity of electrons in a cylindrical conductor is directly proportional to E^2 where E is the electric field applied along the axis of the cylinder. For this kind of material, which of the following graphs represents the correct relation between the current through the material (I) and voltage applied across the material (V)?

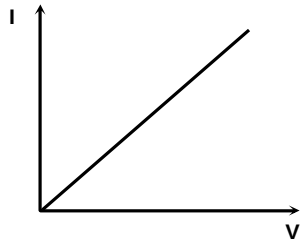
(A)



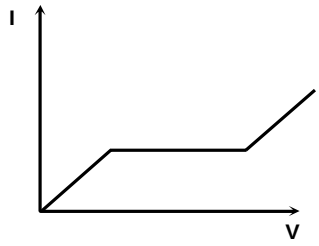
(B)



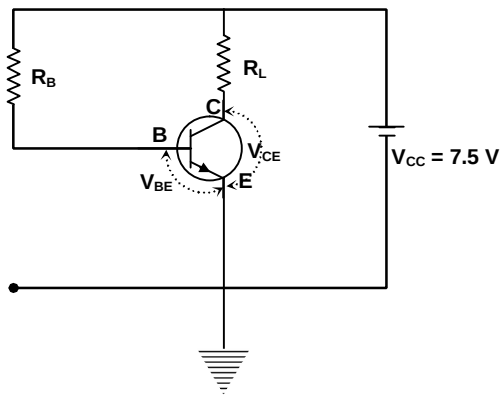
(C)



(D)



10. An n-p-n transistor in a common emitter mode is used as a simple voltage amplifier. The terminal of a 7.5 V battery is connected to the collector through a load resistance R_L and to the base through a resistance R_B = 130 kΩ as shown. If the base-emitter voltage V_{BE} = 1V, the collector-emitter voltage V_{CE} = 3.5 V and β = 100. Then the value of load resistance R_L is



(A) 500 Ω

(B) 600 Ω

(C) 700 Ω

(D) 800 Ω

11. A satellite is following an elliptical orbit around earth. Its engines are capable of providing a one time impulse of a fixed magnitude. In order to maximize energy of satellite the impulse should be given
(A) Along satellite's velocity when satellite is at perigee
(B) Along satellite's velocity when satellite is at apogee
(C) Away from earth, when satellite is at perigee
(D) Away from earth, when satellite is at apogee

12. An unpolarized light of intensity 25 W/m^2 is passed normally through two polaroids placed parallel to each other with their transmission axes making an angle 37° with each other. Then the intensity of the emerging light is
(A) 2 W/m^2
(B) 4 W/m^2
(C) 8 W/m^2
(D) 16 W/m^2

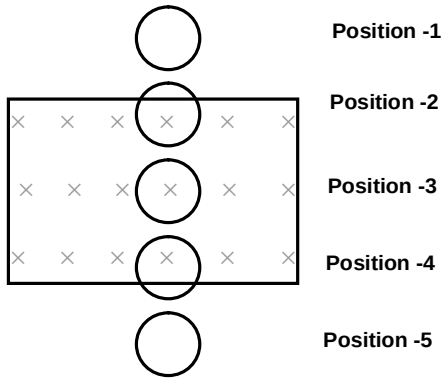
13. Electric potential at the centre of a cube having uniform volumetric charge density 'ρ' and side 'a' is given by V_0. Then, electric potential at the tip of a square pyramid having same volumetric charge density (ρ), of base area $4a^2$ and height 'a' is:
(A) $\frac{V_0}{4}$
(B) $\frac{V_0}{6}$
(C) $\frac{2V_0}{3}$
(D) can't be determined

14. Consider two objects with equal heat capacities and initial temperatures T_1 and T_2. A Carnot engine is run using these two objects as its hot and cold reservoir until they are at equal temperatures. Assume that the temperature change of both hot and cold reservoirs are very small compared to temperature change of Carnot engine during any one cycle. Which option is most appropriate value of final temperature?
(A) $\frac{T_1 + T_2}{2}$
(B) $\sqrt{T_1 T_2}$
(C) $\frac{(T_1 - T_2)}{2}$
(D) $\sqrt{\frac{T_1^3}{T_2}}$

15. A fully discharged battery (emf = 12 V, capacity 60 Ah) is connected with a charging device that can supply the battery a steady current i = 3A. Battery is connected to charging device for 90 minutes. Final charging level of battery (in percentage) is:
(A) 7.5%
(B) 10%
(C) 75%
(D) 15%

16. In the figure shown, there is a constant magnetic field in a rectangular region of space. The field is directed perpendicular into page. A conducting ring moves through region from position 1 to position 5. Induced current is zero when the ring passes through position.

(A) 1 and 5 only
(B) 2 only
(C) 1, 3 and 5 only
(D) 2 and 3 only



17. Number of photons emitted by 10 W sodium vapour lamp in 10 second is : (Assume that 60% of energy is converted into light. Wavelength of sodium light = 390 nm)

(A) 1.18×10^{20}
(B) 1.77×10^{18}
(C) 3.4×10^{19}
(D) 3.4×10^{18}

18. A neutron collides elastically with initially stationary deuteron. The fraction of kinetic energy lost by neutron is (assume head on collision)

(A) $\frac{8}{9}$
(B) 1
(C) $\frac{2}{3}$
(D) $\frac{1}{2}$

19. An object of mass m is projected with a momentum p at an angle with the horizontal such that its maximum height (H) is half of its Range (R). Minimum kinetic energy of the particle in its path will be

(A) $\frac{p^2}{8m}$
(B) $\frac{3p^2}{4m}$
(C) $\frac{p^2}{10m}$
(D) $\frac{p^2}{5m}$

20. A particle of mass 2 kg is executing simple harmonic motion with amplitude A_0. At t = 0, a small resistive force also starts acting on the particle which is given by $\vec{F} = \frac{-\vec{v}k\ell n2}{10}$. The value of amplitude of the particle after 20 second will be: (where k = 1 kg/sec)

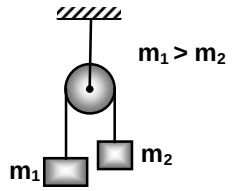
(A) $A_0/2$
(B) $A_0/4$
(C) A_0
(D) $A_0/\sqrt{2}$

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. The coercive force for certain permanent magnet is 2×10^4 A/m. This magnet is placed inside a long solenoid of 40 turns/cm and a current is passed in the solenoid to demagnetize the magnet. The value of current (in A) is

22. Pulley of atwood machine has mass m_1 and is free to rotate about its geometrical center without friction. Centre of mass of the pulley is at a distance R/2 from geometrical centre, where R is the radius of the pulley. There is no slipping between the string and pulley. The ratio of maximum acceleration to minimum acceleration of block of mass m_1 is



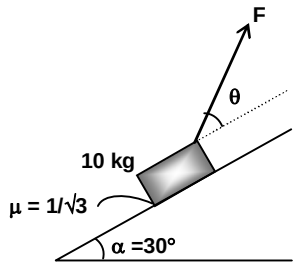
(Take m_1R^2 as moment of inertia of pulley about axis of rotation, $\frac{m_1}{m_2} = 3$)

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. In a YDSE experiment, one of the slits is covered with a glass plate of thickness (2.4×10^{-4}) meter refractive index of which varies with time as $\mu_S = t^2 + 12t + 8$. The separation between the slits is 2×10^{-3} meter and the distance between the plane of slits and the screen is 2 meters. Find the velocity (in m/s) of the central maxima at t = 4 sec. (Take refractive index of air = 1)

24. A block of mass 10 kg is kept on an inclined plane of inclination 30° and friction coefficient between the block and inclined plane is $\mu = \frac{1}{\sqrt{3}}$. Find the minimum value of force (in newton) required to move the block up the inclined plane.
(Take $\sqrt{3} = 1.732$, g = 10 m/s^2)



25. In a resonance tube experiment, 3rd resonance occurs at 111 cm and 5th resonance occurs at 200 cm. Find the end correction (in cm).

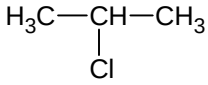
Chemistry

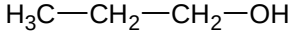
PART – II

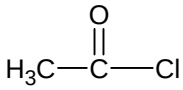
SECTION – A
(One Options Correct Type)

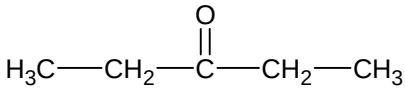
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. Among the following, which one give iodoform test?

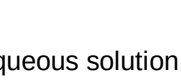
(A) 

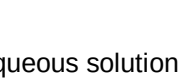
(B) 

(C) 

(D) 

27. Which one of the following solution behave as buffer solution?

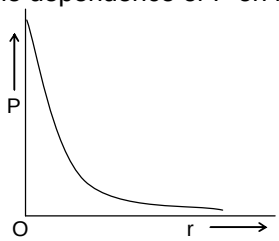
(A) Aqueous solution of  and  .

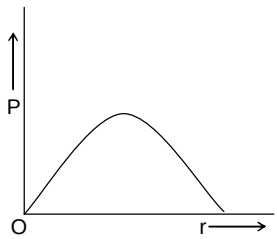
(B) Aqueous solution of  and HCl.

(C) Aqueous solution of NH_4OH and NaOH.

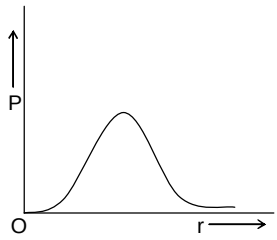
(D) Aqueous NaCl solution.

28. P is the probability of finding the 2s electron of hydrogen atom in a spherical shell of infinitesimal thickness, dr at a distance r from nucleus, the volume of this shell is $4\pi r^2 dr$. The qualitative sketch of the dependence of P on r is

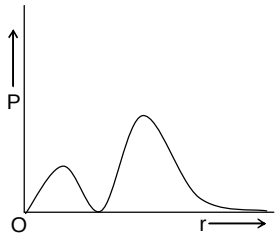
(A) 

(B) 

(C)



(D)



29. Among the following, intensive property is
(A) E_{cell}
(B) Mass
(C) Internal energy
(D) Heat capacity

30. A diatomic ideal gas undergoes a process in which the ratio of P to V^2 at any instant is constant and equals to 1. What is molar heat capacity of gas [excluding vibrational degree of freedom]
(A) 2R
(B) $\frac{5R}{2}$
(C) $\frac{17R}{6}$
(D) $\frac{4R}{3}$

31. Which one of the following is orthosilicate?
(A) Forsterite
(B) Hemimorphite
(C) Barysilite
(D) Thortveitite

32. Number of coordination isomers possible for the given complex $\left[\overset{III}{Co}(en)_3\right]\left[\overset{III}{Cr}(CN)_6\right]$ is (including itself)
(A) 6
(B) 4
(C) 2
(D) Not possible.

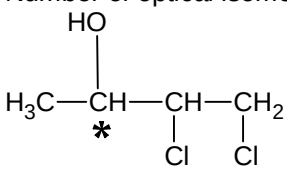
33. Number of stereo isomers possible for the given complex $[Ma_2b_2cd]$ [Where a, b, c, and d are monodentate ligand] is
(A) 8
(B) 6
(C) 5
(D) 7

34. Among the following, the incorrect statement is
(A) P_4O_{10} is an oxidizing agent
(B) P_4O_{10} is simple anhydride
(C) P_4O_{10} at high temperature react with sodium chloride to form phosphoryl chloride
(D) P_4O_{10} is a dehydrating agent.

35. The number of minimum hydrogen atom in sample to get maximum number of different spectral lines for the transition n = 4 to n = 1 is
(A) 4
(B) 6
(C) 10
(D) 1

36. Among the following, sulphate ore is
(A) Zinc blend
(B) Barytes
(C) Pitch blend
(D) Dolomite

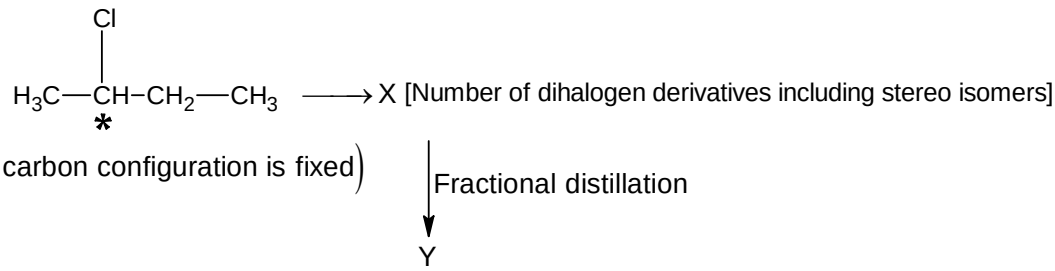
37. Number of optical isomers possible for the given organic molecule



is

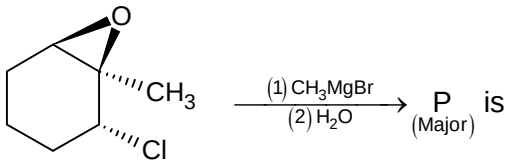
[at '$\overset{*}{C}$' carbon configuration is fixed]
(A) 4
(B) 3
(C) 2
(D) 6

38.

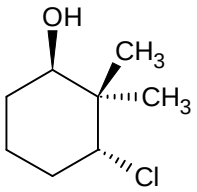


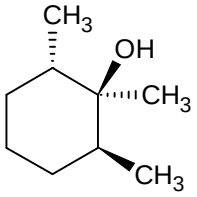
Then the value of (X + Y) is
(A) 10
(B) 9
(C) 4
(D) 8

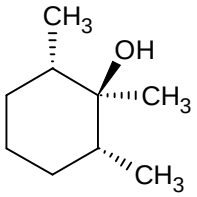
39.

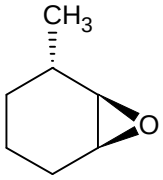


is

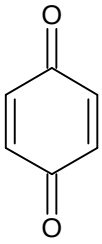
(A) 

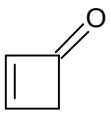
(B) 

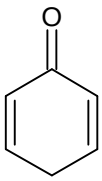
(C) 

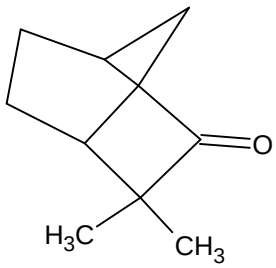
(D) 

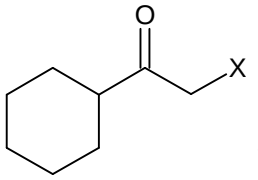
40. Among the following, in which tautomerism is possible?

(A) 

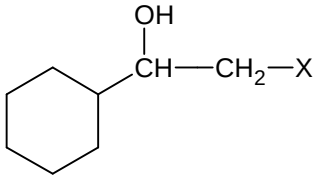
(B) 

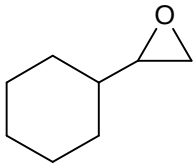
(C) 

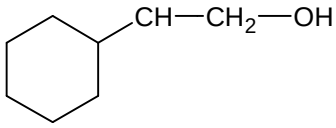
(D) 

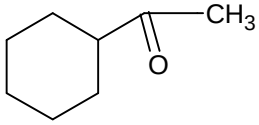
41. 

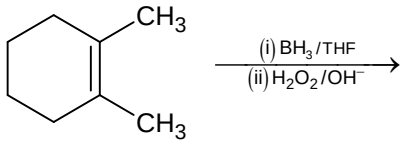
$\xrightarrow{NaBH_4}$ P(Major) is

(A) 

(B) 

(C) 

(D) 

42. 

Among the following correct statement is:

(A) Number of product in given reaction is one.
(B) Reaction is regioselective.
(C) Number of products in given reaction is four.
(D) Number of products in given reaction is two.

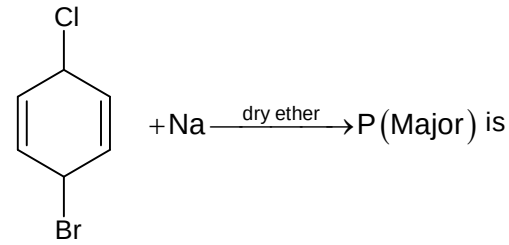
43. The equation of state for a given gas is P(V – nb) = nRT, where b & R are constant, if the pressure and temperature are such that V_m = 11b, then what is the compressibility factor of gas in this condition:

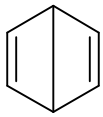
(A) $\frac{10}{9}$

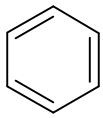
(B) $\frac{9}{11}$

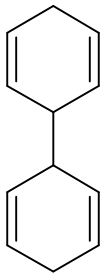
(C) $\frac{11}{10}$

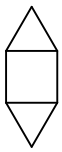
(D) $\frac{10}{11}$

44. 

(A) 

(B) 

(C) 

(D) 

45. The rate of reaction may be expressed as
$\frac{1}{4}\frac{d(C)}{dt} = \frac{-1}{3}\frac{d(D)}{dt} = \frac{1}{4}\frac{d(A)}{dt} = \frac{-d(B)}{dt}$, the reaction is

(A) $4A + B \longrightarrow 2C + 3D$

(B) $B + 3D \longrightarrow 4A + 4C$

(C) $4A + 2C \longrightarrow B + 3D$

(D) $2C + 3D \longrightarrow 4A + 2B$

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. Bond order of Be_2^+ is x, then what is the value of 2x:

47. Number of dibasic acid in the following oxoacid is
H_3BO_3, H_2SO_4, H_2SO_5, H_3PO_3, H_3PO_4, $H_2S_2O_7$.

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

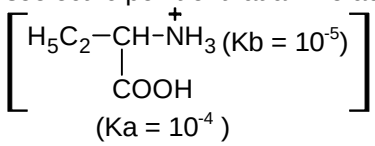
48. Pressure over an ideal binary liquid solution containing 10 moles of each of liquid A and B is gradually decreased isothermally. At what pressure (in torr), half of the total amount of liquid will get converted into vapour? $\left(P_A^o = 400 \text{ torr}, P_B^o = 100 \text{ torr}\right)$

49. For the given redox reaction

$$K_2Cr_2O_7 + 14HCl \longrightarrow 2KCl + 2CrCl_3 + 3Cl_2 + 7H_2O$$

What is the equivalent mass of HCl? [Given : M_{HCl} = 36.6 g/mol]

50. In an amino acid the carboxylic acid group has $K_a=10^{-4}$ and amino group has $K_b= 10^{-5}$. pH at isoelectric point of that amino acid is



Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. If y = f(x) is monotonic function in (–3, 9), then the area bounded by the ordinates x = –3, x = 9, y = f(x) and y = f(c) where c $\in$ (–3, 9) is minimum when c is equal to
(A) 0
(B) 3
(C) 6
(D) none of these

52. The digit at hundredths place in coefficient of x^{22} in the expansion of $(1 – x^7 + x^8)^{25}$ is equal to
(A) 0
(B) 9
(C) 6
(D) 2

53. If $a^{2x} – 14a^x + 53 + | |b + 3| – 4| = |4 \cos \theta|$ where x, a, b and $\theta \in R$, then the sum of the possible values of b is equal to
(A) –6
(B) 6
(C) 4
(D) –4

54. If $\vec{r} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{r}_1 = \left(a\hat{i} + b\hat{j} + c\hat{k}\right)$ where a, b, c $\in$ {–3, –2, –1, 0, 1, 2, 3, 4}, then number of possible non-zero vector $\vec{r}_1$ perpendicular to $\vec{r}$ is
(A) 45
(B) 55
(C) 46
(D) none of these

55. Let $f(x) = x + x^3 + x^7 + x^{12} + x^{19} + \ldots..$ the coefficient of x^{15} in f(f(x)) is p, then number of even divisor of p is equal to
(A) 10
(B) 8
(C) 18
(D) 80

56. If A is a square matrix of order 2, and $(tr(A))^2 – tr(A^2) = k \det(A)$, where k lies between roots of equation $x^2 – ax + (a – 1) = 0$, then least integral value of a is
(A) 3
(B) 4
(C) 5
(D) 2

57. Let matrix $A = \begin{bmatrix} x & y & 2 \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$ where $x, y \in N$. If $|adj(adj(adj(adj\ A)))| = 3^{32} \cdot 5^{16}$, then number of such matrix A is equal to
(A) 46
(B) 47
(C) 48
(D) none of these

58. Let $f(x) = (x^2 - 5x + 4)(x^2 + 5x + 4)$ and α, β, γ (where $\alpha < \beta < \gamma$) are roots of $f'(x) = 0$, then the value of $\frac{[\alpha]^2 + [\beta + 1]}{[\gamma]}$ is equal to (where [.] denote greatest integer function)
(A) 0
(B) $\frac{5}{3}$
(C) $\frac{5}{2}$
(D) 5

59. Let $f(x) = \max\{\cos x, x, x^2\}$ in $-3 \le x \le 3$, then
(A) f(x) is continuous everywhere but not differentiable at exactly 3 points
(B) f(x) is continuous everywhere but not differentiable at exactly 2 points
(C) f(x) is not differentiable at four points
(D) none of these

60. If $a, b, c \in R^+$ such that $a + b + c = 27$ and maximum value of $a^2 b^3 c^4 = 2^m \times 3^n$, then sum of six Am's between m and n is
(A) 66
(B) 132
(C) 120
(D) 176

61. In a ΔABC vertex A and B lies on x-axis and y-axis respectively, where A(a, 0) is a fixed point, B is a variable point such that $\angle C = \tan^{-1}\frac{4}{3}$ and AC = BC, then locus of C
(A) $2x - 4y + 3a = 0$
(B) $4x + 2y - 3a = 0$
(C) $2x + 4y + 7a = 0$
(D) $4x - 2y + 5a = 0$

62. If ℓ_1 and ℓ_2 are lengths of two perpendicular focal chord of parabola $y^2 = 16(x + 1)$, then harmonic mean of ℓ_1 and ℓ_2 is
(A) 4
(B) 8
(C) 16
(D) 32

63. Eccentricity of curve represented by $8x^2 + 4\sqrt{6}xy + 6y^2 = 5$
(A) $\sqrt{\frac{5}{6}}$

(B) $\sqrt{\frac{7}{6}}$

(C) $\frac{1}{\sqrt{6}}$

(D) none of these

64. Let P(x) is polynomial of degree four with leading coefficient one and satisfying the condition P(3) = 11; P(–3) = 11, P′(3) = 6, P″(3) = 2, then value of P(5) is equal to
(A) 75
(B) 87
(C) 27
(D) 91

65. Locus of image of point (–2, 3) about the line (x + y + 7) + k(x + 2y + 9) = 0 (k is non-zero real numbers is
(A) straight line
(B) circle with radius = $\sqrt{34}$
(C) ellipse whose $e = \frac{2}{3}$
(D) hyperbola whose $e = \sqrt{2}$

66. A dice is constructed such that probability of occurrence of a number is proportional to the square of number, then probability that three occur at least once when dice throw 10 times is equal to
(A) $^{10}C_1 \times \left(\frac{82}{91}\right)^9 \left(\frac{9}{91}\right)$
(B) $\left(\frac{82}{91}\right)^{10}$
(C) $\frac{91^{10} - 82^{10}}{91^{10}}$
(D) none of these

67. Let p and q be two statements, then $(\sim p \vee q) \wedge (\sim p \wedge \sim q)$ is a
(A) tautology
(B) contradiction
(C) neither tautology nor contradiction
(D) none of these

68. The variance of α, β, γ and δ is 9, then variance of 7α, 7β, 7γ and 7δ is
(A) 63
(B) $\frac{9}{7}$
(C) $\frac{7}{9}$
(D) 441

69. A function f(x) satisfying $\int_0^1 f(tx)\,dt = nf(x)$, where x > 0, is
(A) $f(x) = c \cdot x^{\left(\frac{1-n}{n}\right)}$

(B) $f(x) = c \cdot x^{\left(\frac{n}{n-1}\right)}$

(C) $f(x) = c \cdot x^{\left(\frac{1}{n}\right)}$

(D) $f(x) = c \cdot x^{(1-n)}$

70. Equation of image of the line $x + y = \sin^{-1}(\lambda^6 + 1) + 5\cos^{-1}(\lambda^4 + 1) - \tan^{-1}(\lambda^2 + 1)$, where $\lambda \in R$ about x-axis is given by

(A) $x - y = 0$

(B) $x - y = \frac{\pi}{2}$

(C) $x - y = \pi$

(D) $x - y = \frac{\pi}{4}$

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. If $\int \frac{\text{cosec}^2 x - 2020}{\cos^{2020} x} dx = \frac{Af(x)^B}{(g(x))^{2020}} + c$; where $f\left(\frac{\pi}{6}\right) = \sqrt{3}$, then value of $A^2 + B^2 + f\left(-\frac{\pi}{4}\right)$ is equal to

72. The minimum value of $\frac{r_1 r_2 r_3}{(3r)^3}$ in a triangle is (where r is in radius and r_1, r_2, r_3 are exradii)

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. Angle of intersection between the curve $\frac{x^2}{36} + \frac{y^2}{4} = 1$ and $x^2 + y^2 = 12$ is $\theta = \sin^{-1}\left(\frac{b}{\sqrt{a}}\right)$, then the value of $\frac{a}{2b}$ is equal to

74. Image of point (3, 2, 0) about the plane $x + 2y + z + 1 = 0$ lies on plane $\lambda x + y - 3\lambda z + 7 = 0$, then $|\lambda|$ is

75. Number of words that can be formed using letters of word CAULIFLOWERS in which vowels are in alphabetic order is m and the words in which no two vowels are together is n, then $\frac{14m}{n}$ is equal to